JOM, Vol. 78, No. 1, 2026
https://doi.org/10.1007/s11837-025-07619-1
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Electrodeposition of Ni-Mo-P Ternary Alloy Electrocatalyst for Hydrogen Evolution Reactions

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Alkaline water electrolysis is an important method for sustainable hydrogen production, and exploring efficient cathode materials is of utmost importance. In this study, Ni-Mo-P/NF hydrogen evolution cathode materials for alkaline water electrolysis-based hydrogen production were fabricated by electrodeposition method. During this process, with nickel foam (NF) as the substrate, the composition of the plating solution is nickel sulfate, sodium molybdate and sodium hypophosphite. The ions of Ni^{2+}, MoO_4^{2-} and $H_2PO_2^-$ in the plating solution were reduced and co-deposited on the surface of the foamed nickel, forming a Ni-Mo-P alloy coating. Furthermore, the electrode surface presented a honeycomb-like structure. The Ni-Mo-P/NF cathode electrode material exhibited excellent electrocatalytic performance, as demonstrated by electrochemical performance tests. In a 1 M KOH electrolyte, the hydrogen evolution reaction (HER) has a relatively low overpotential η_{10} of 0.109 V, with a Tafel slope of 121.8 mV/dec; the AC impedance R_{ct} is 56.47 Ω; the electrochemical specific surface area is 87.6 cm^2. This outstanding catalytic activity is attributed to the synergistic effect among Ni, Mo and P. Mo modulates the electronic structure of Ni, enhancing the adsorption and dissociation of water molecules, whereas P increases the overall catalytic efficiency of the material. This study provides valuable references for the development of highly active nonmetal-based cathode materials in the alkaline water electrolysis field

INTRODUCTION

In the present era, the excessive exploitation and utilization of the limited and non-renewable fossil energy resources have reshaped the global energy landscape. It is urgent to adjust the energy structure, reduce the dependence on high-carbon fossil energy and vigorously promote the transformation of the energy system in a low-carbon and green direction.[1–3] Green hydrogen obtained through photocatalysis[4–7] and electrolysis of water[8–10] has become a research hotspot due to its high purity and zero pollution characteristics.

Although the photocatalytic water splitting for hydrogen production technology has great potential, its efficiency is generally low, usually ranging from 10% to 20%.[11] For hydrogen production through water electrolysis, proton exchange membrane electrolysis (PEM) and solid oxide electrolysis (SOEC) have efficiencies as high as 90%, but they are limited by the high costs and immature technologies, making it difficult to achieve large-scale industrialization applications in a short period of time.[12,13] Compared with the three hydrogen production technologies mentioned above, the technology of alkaline water electrolysis (AWE) for hydrogen production is mature, and the lower cost is suitable for large-scale commercial

(Received March 25, 2025; accepted July 9, 2025; published online August 21, 2025)

applications.[14–17] However, it also encounters some bottleneck problems, such as lower hydrogen production efficiency, which needs to be broken.

Precious metal electrode materials such as Pt, Pd, Ru and Rh have good hydrogen evolution reaction activity, which can effectively reduce the overpotential of hydrogen evolution and promote the occurrence of hydrogen evolution reaction in the process of water electrolysis. However, owing to its high price and low abundance, the production cost cannot be effectively reduced, so it is not conducive to large-scale production.[18–20] Therefore, the development of high-performance and low-cost hydrogen evolution cathode materials has become a key problem. Currently, the three main factors that affect the electrochemical performance of the catalyst can be further improved through modification treatment, including: exposing more active sites and increasing the number of active sites, enhancing the conductivity of the catalyst and improving the rate of mass transfer, optimizing the adsorption energy of the catalytic material's active sites for the intermediate products of the hydrogen evolution reaction and enhancing the site activity. To address the aforementioned challenges, researchers have proposed several effective strategies, including elemental doping,[21,22] defect engineering[23] and structural optimization.[24,25] Additionally, morphology engineering of the catalytic materials has proven effective in achieving these objectives.[26–29]

At present, transition metal phosphides have excellent hydrogen evolution performance in alkaline electrolytes. As new members in the field of hydrogen evolution, transition metal phosphides have attracted much attention because of their unique structure of approximately spherical triangular prisms and the properties of metal conductors. Its unique geometric structure can expose more coordinated unsaturated surface atoms and provide more active sites for electrocatalysis.[30–33] Shaojun Sun[34] conducted two steps: (1) pretreatment and in-situ hydrolysis; (2) synthesis of the $Ni_5P_4/Ni_2P/Fe_2P$ electrocatalyst via low-temperature phosphorylation. The overpotential of the hydrogen evolution reaction was 190 mV at a current density of 10 $mA{\cdot}cm^{-2}$. Yanhong Wang[35] constructed a series of M (Fe, Co, Cu, and Cr)-doped M-Ni_5P_4-Moh with abundant oxygen positions via hydrothermal methods and partial phosphatization methods, and the hydrogen evolution cathode overpotential η_{10} of 157 mV can be obtained.

According to the above analysis, the TMPs showed high activity in alkaline electrolytic water. However, many aspects can be further improved through processing to increase the electrocatalytic activity. The three main factors influencing its electrochemical performance include: exposing more active sites and increasing the number of active sites, enhancing the conductivity of the catalyst and improving the rate of mass transfer, optimizing the adsorption energy of the catalytic material's active sites for the intermediate products of the hydrogen evolution reaction and the HER and enhancing the site activity. Therefore, in this work, we propose the preparation of a Ni-Mo-P/NF porous hydrogen evolution cathode with nickel sulfate, sodium molybdate and sodium hypophosphite as the electrolyte components via direct current electrodeposition technology and nickel foam as the base. Taking overpotential as the index, the catalytic effects of Na_2MoO_4, NaH_2PO_2 and $NiSO_4$ concentrations on the hydrogen evolution cathode catalyst Ni-Mo-P/NF were investigated, and the optimal process parameters were obtained. The microstructures of the materials were characterized via SEM and XRD. The electrochemical performance will be evaluated through overpotential, cyclic voltammetry, double-layer capacitance, turnover frequency and impedance. The hydrogen evolution performance will be assessed accordingly.

Hydrogen Evolution Reaction

For the hydrogen production process of AWE, water splitting is conducted with hydrogen evolution reaction (HER) at the cathode to obtain hydrogen and an oxygen evolution reaction (OER) at the anode to produce oxygen.

The hydrogen evolution reaction at the cathode is a two-electron reaction process, and its stepwise reaction process is shown in Eq. 1 and Eq. 3. The Volmer step determines the adsorption process of hydrogen ions, whereas the Tafel and Heyrovsky steps are the desorption processes of hydrogen gas and are controlled by the Tafel and Heyrovsky steps. Therefore, the hydrogen evolution reaction process always involves the Volmer adsorption step, and according to the difference in the desorption step, the alkaline electrolytic water splitting process for hydrogen production can be classified into the Volmer-Tafel or Volmer-Heyrovsky reaction pathways. In the hydrogen evolution reaction process, the Tafel slope is usually used to determine the reaction kinetics. The smaller the Tafel slope is, the faster the reaction rate. When the Tafel slope is < 30 mV dec^{-1}, the rate-controlling step of the hydrogen evolution reaction is the Tafel reaction; when the value of Tafel slope is in the range of 40 mV $dec^{-1} \sim 120$ mV dec^{-1}, the rate-determining step of the hydrogen evolution reaction is the Heyrovsky reaction; when the Tafel slope is > 120 mV dec^{-1}, the rate-determining step of the hydrogen evolution reaction is the Volmer reaction.

Volmer step:

$$^{*} + H_2O + e^- \rightarrow H^{*} + OH^- \quad (1)$$

Tafel step:

$$H^{*} + H^{*} = H_2 \quad (2)$$

Heyrovsky step:

$$H^* + H_2O + e^- = H_2 + OH^- \quad (3)$$

Among them, * represents the reaction site, and H^* is the intermediate of hydrogen adsorption.

Besides, a simulation model for the Ni-Mo-P catalyst was established, and DFT theoretical calculations were conducted on the elementary reactions of each step of the hydrogen evolution reaction to reveal the reaction mechanism and promote the synergistic effect of the ternary coating on the catalytic activity of the hydrogen evolution electrode.

EXPERIMENTAL DESIGN AND SIMULATION CALCULATION

Characterization of the Morphology, Composition and Structure of the Electrode

The morphology and elemental distribution of the coating surface were analyzed using a ZEISS Gemini 300 scanning electron microscope (SEM) and energy-dispersive spectrometer (EDS). Elemental distribution electrodeposition was characterized via x-ray diffraction (XRD, Smart Lab) with a 2θ scanning angle ranging from 10° to 90°. The target material was a closed target (Cu, Ka), and the test speed was 5°/min.

Electrochemical Tests

A CHI760E electrochemical workstation equipped with a standard three-electrode system was applied to investigate the hydrogen evolution catalytic performance of the hydrogen evolution electrode. The auxiliary electrode was a platinum electrode, which could form a circuit with the working electrode to ensure the unobstructed current through the working electrode, thereby guaranteeing that the researched reaction occurred on the electrode and achieved its polarization.[20] The reference electrode was a saturated calomel electrode (SCE), and the solution inside the calomel electrode was usually a saturated KCl solution. The prepared Ni-Mo-P/NF was used as the working electrode. According to the formula $E_{\mathrm{vs.RHE}} = \mathrm{E}_{\mathrm{vs.SCE}} + 0.059 \times \mathrm{pH} + 0.2415\mathrm{V}$, the measured values obtained from the SCE electrode can be converted into the standard reversible hydrogen electrode potential (RHE).

Linear sweep voltammetry is the I-E curve (LSV) where the current varies with the potential.[20] The electrocatalytic performance of different electrodes can be analyzed by comparing their linear sweep voltammetry curves in the same alkaline solution.

The overpotential for hydrogen evolution obtained from these curves can be adopted to evaluate the electrocatalytic performance of the electrodes. The overpotential is an important indicator for evaluating catalytic reactions. It is usually defined as the voltage corresponding to a current density of 10 mA/cm^2. A smaller overpotential value means better the catalytic activity.[20] The scanning range of the linear scanning voltammetry experiments was set to the open-circuit voltage of the electrode (OCP) ± 0.3 V, and the scanning speed was 1 mV/s.

The Tafel curve conforms to the Tafel relationship. The corresponding current is expressed in the form of the logarithm of its absolute value to obtain the log|i| - E graph. In this graph, the "Tafel zone" shows a linear relationship, which can be fitted to obtain the following equation:

$$\eta = \mathrm{a} + b \log j \quad (4)$$

In the formula: η = overpotential

a = 0verpotential for hydrogen evolution at an electric current density of i = 1 A/cm2, constant

b = Tafel slope, which is the slope of the straight line in the log|i| - E graph, is a constant.

j = current density

The constant value of a is affected by the experimental temperature and the concentration of the plating solution for samples preparation and the surface state of the prepared samples. The current density j and the Tafel slope b are related to the electrode material and the reaction mechanism of the electrode, respectively. Generally, the larger the current density j of the electrode, the smaller the Tafel slope b, which proves that the catalytic hydrogen evolution performance of the electrode is better.

The impedance data of the real part and imaginary part were obtained through the exchange impedance method.[37] By fitting the equivalent circuit via ZSimpWin software, the solution resistance R_s, the electrochemical reaction resistance R_{ct} and the double-layer capacitance Cdl were obtained. If the drawn graph takes on a semicircular shape, the Volmer-Heyrovsky hydrogen evolution reaction has occurred on the surface of the prepared samples. If it is a semicircle combined with a straight line, it can be concluded that the hydrogen evolution reaction is of the Volmer-Tafel type.[36] The semicircular diameter represents the magnitude of the electrochemical reaction impedance R_{ct}, which can indicate the impedance of the electrode reaction. In the experiment, the frequency adopted for the alloy electrode ranged from 0.01 Hz to 100 kHz.

To evaluate the influence of the actual surface area of the reaction on the performance of the hydrogen evolution reaction, the effective electrode surface area (ECSA) of the cathode was evaluated via the double-layer capacitance method.[37] It determines the number of active sites exposed during the reaction, and a cathode with a larger ECSA indicates better catalytic activity. The electrochemical double-layer capacitance (Cdl) of each sample was measured via cyclic voltammetry (CV). The data range was set to OCP ± 0.05 V. A series of CV curves was obtained by varying the scan rate (10, 20, 30, 40, 50, 60 mV/s). The slope value of the fitted

curve is twice that of Cdl and is in direct proportion to the ECSA value, as shown in Eq. 5 below.

$$ECSA = \frac{C_{dl}}{C_s} \quad (5)$$

where C_{dl} is the double-layer capacitance value.

C_s is the specific capacitance value: 0.040 mF/cm^2.

Turnover Frequency TOF

The TOF (transition frequency of the active site) value can be adopted to evaluate the intrinsic catalytic activity conversion frequency value of each catalytic active site. The TOF is defined as the number of catalytic reactions occurring per unit time and per unit active site under certain reaction conditions or the quantity of the target product generated or the quantity of reactants consumed.[38]

$$TOF = \frac{jA}{znF} \quad (6)$$

In the formula: j = current density under overpotential

A = working electrode area

Z = number of electron transfers in catalytic reactions

N = amount of the substance at the active site

F = Faraday constant (96485 C·mol^{-1})

Optimization of the Electrodeposition Process

Preparation of the Ni-Mo-P/NF Porous Hydrogen Evolution Cathode

The substrate used in the experiment was nickel foam (NF) (99.99%, thickness 1.5 mm). Before application in the electrodeposition process, it was retreated in an ethanol medium with ultrasonic assistance to remove impurities, such as oil stains and oxide films on the surface and in the pores of the nickel foam.

The Ni-Mo-P alloy electrode was fabricated via the constant current electrodeposition method. The composition of the electrolyte was composed of $NiSO_4{\cdot}6H_2O$, $Na_2MoO_4{\cdot}2H_2O$, $NaH_2PO_2{\cdot}H_2O$, $Na_3C_6H_5O_7{\cdot}2H_2O$, $NH_3{\cdot}H_2O$ and NaCl. The anode and cathode of the experimental device are nickel foam and graphite, respectively. The three electrodes are fixed and inserted into a beaker, and the bottom of the electrodes is 3 cm below the interface of the electrolyte and placed on a temperature-controlled magnetic stirrer. The electroplating process was conducted under the condition of deposition temperature 55°C, pH 10, and stirring speed 300 rpm/min. Information regarding the material suppliers is presented in Table I.

The overpotential for hydrogen evolution of the coating reflects the excellent catalytic hydrogen evolution performance of the coating. Here, the overpotential for hydrogen evolution is applied as the evaluation index to optimize the electrodeposition process.

Experimental Optimization of the Electrodeposition Process

The preliminary orthogonal experiments showed that the concentrations of sodium molybdate, nickel sulfate and sodium hypophosphite have significant influences on the electrochemical performance of Ni-Mo-P/NF electrode materials. To obtain the optimal process parameters, single-factor experiments were designed. The concentration of Na_2MoO_4, NaH_2PO_2 and $NiSO_4$ was controlled within the range of 0.12 – 0.20 mol/L, 0.16 – 0.24 mol/L and 0.12 – 0.20 mol/L, respectively. The overpotential is adopted as an indicator to measure the catalytic activity of the Ni-Mo-P/NF hydrogen evolution cathode. The preparation scheme is shown in Fig. 1.

Figure 2a shows the LSV curves of the Ni-Mo-P ternary alloy coatings electroplated with different sodium molybdate concentrations. With increasing Na_2MoO_4 concentration in the plating solution from 0.12 mol/L to 0.16 mol/L, the hydrogen evolution catalytic activity of the porous electrode is significantly enhanced. However, if the concentration of sodium molybdate is further increased, the hydrogen evolution catalytic activity of the electrode decreases instead. This may be because excessive MoO_4^{2-} in the plating solution hinders the electrodeposition process and reduces the hydrogen evolution activity of the coating layer. A comparison of the LSV detection data revealed that when the concentration of Na_2MoO_4 was 0.16 mol/L, an electrode with better performance could be prepared.

Figure 3a shows the linear sweep voltammograms of the Ni-Mo-P/NF ternary porous electrode with different concentrations of NaH_2PO_2. The smallest hydrogen evolution overpotential η_{10} of the electrodes prepared with NaH_2PO_2 concentration of 0.20 mol/L is 0.144 V, as shown in Fig. 3b. As the concentration of sodium metabisulfite in the plating solution increases, the overpotential for hydrogen evolution first decreases and then increases. The possible reason for this trend may be that the secondary bisulfite ion reaction occurred in the codeposition process of the Ni-Mo-P system, as follows:[39]

$$H_2PO_2^- + H_2O = HPO_3^{2-} + H^+ + 2[H] \quad (7)$$

A portion of the precipitated [H] will reduce Ni^{2+} to Ni, MoO_4^{2-} to Mo and $H_2PO_2^-$ to P as follows:

$$Ni^{2+} + 2[H] = Ni + 2H^+ \quad (8)$$

$$MoO_4^{2-} + 6[H] = Mo + 2OH^- + 2H_2O \quad (9)$$

$$H_2PO^{2-} + [H] = H_2O + OH^- + P \quad (10)$$

Table I. Experimental chemical reagents

Molecular formula	Factory	Grade of purity	Region
$NiSO_4{\cdot}6H_2O$	Tianjin Aopu Sheng Chemical Co., Ltd	AR	China
$Na_2MoO_4{\cdot}2H_2O$	Tianjin Chemical Reagent No. 4 Factory	AR	China
$NaH_2PO_2{\cdot}H_2O$	Sinopharm Chemical Reagent Co., Ltd	AR	China
$C_6H_5Na_3O_7{\cdot}2H_2O$	Tianjin Damao Chemical Reagent Factory	AR	China
NaCl	Tianjin Fuyu Fine Chemical Co., Ltd	AR	China
NH_4OH	Tianjin Damao Chemical Reagent Factory	AR	China

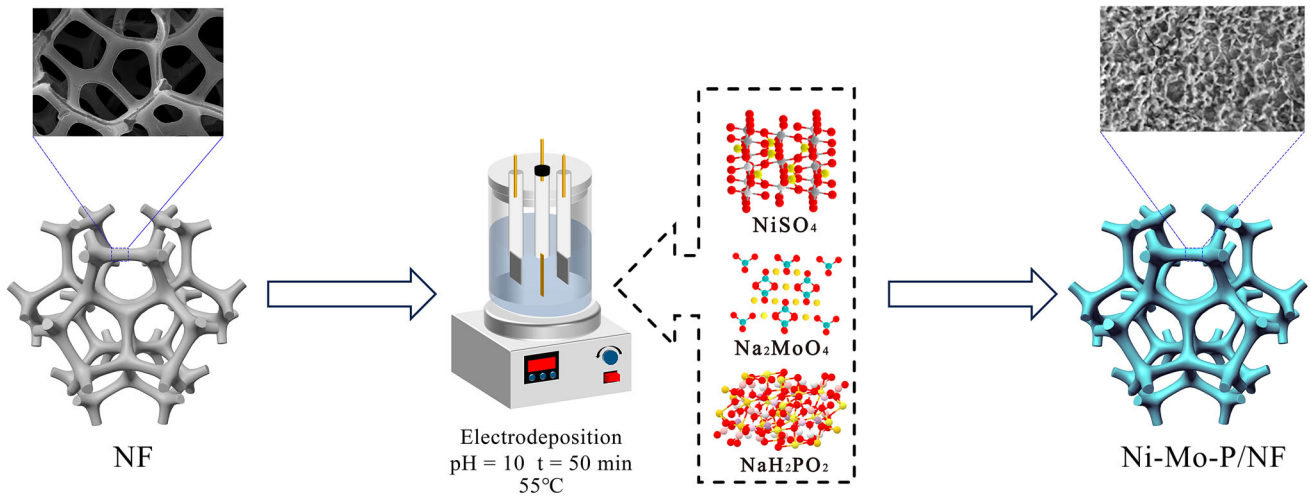

Fig. 1. Preparation of the Ni-Mo-P/NF electrodes via constant current electrodeposition method.

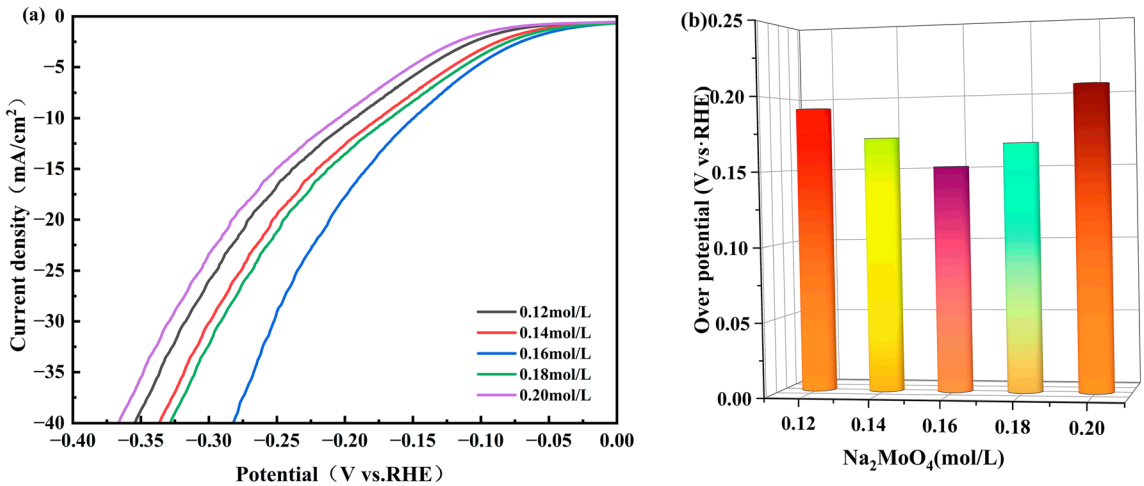

Fig. 2. Effect of the Na_2MoO_4 concentration on the hydrogen evolution performance of Ni-Mo-P/NF: (a) LSV curves; (b) overpotential η_{10}.

For the above possible reactions, the higher the concentration of NaH_2PO_2, the more active [H] will be electrolyzed, and more Ni, Mo and P can be reduced. The catalytic hydrogen evolution activity of the coating layer becomes increasingly enhanced. Data obtained from the LSV detection method comparison showed that when the concentration of NaH_2PO_2 is 0.20 mol/L, an electrode with better performance can be prepared.

Figure 4a shows the linear sweep voltammograms of the Ni-Mo-P/NF ternary porous electrode with different concentrations of $NiSO_4$. As shown in Fig. 4b, with increasing nickel sulfate concentration, the catalytic hydrogen evolution activity first increases but then decreases. When the $NiSO_4$ is 0.16 mol/L, the overpotential is relatively low and the catalytic hydrogen evolution performance is better. This might be because when the nickel source concentration is too high, the supply of P ions

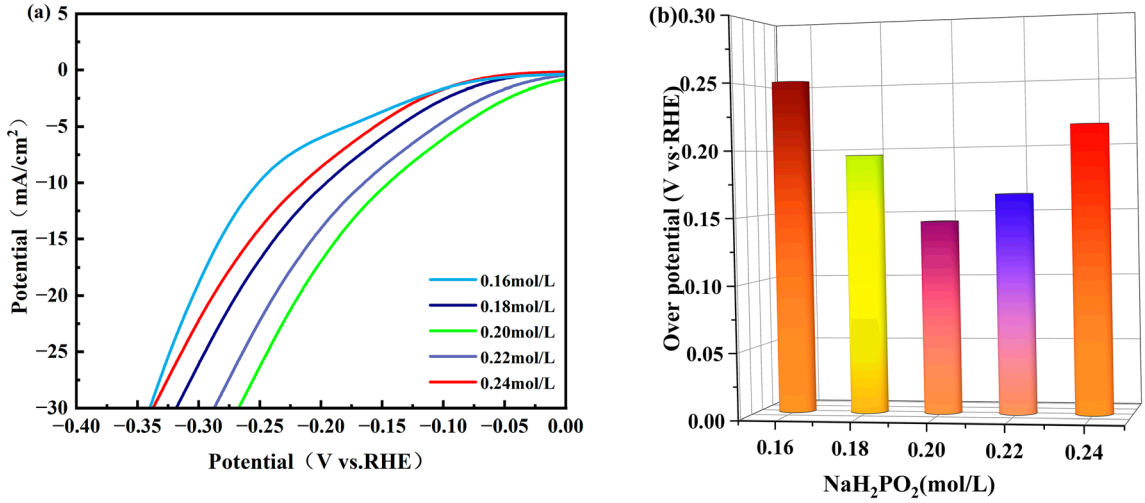

Fig. 3. Effect of the NaH_2PO_2 concentration on the hydrogen evolution performance of Ni-Mo-P/NF: (a) LSV curves; (b) overpotential η_{10}.

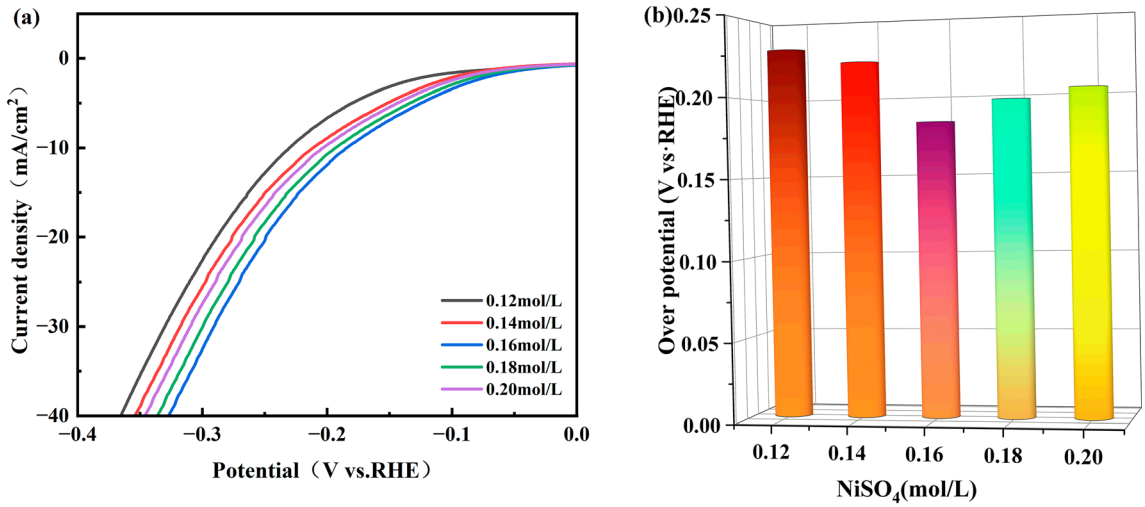

Fig. 4. Effect of $NiSiO_4$ concentration on hydrogen evolution performance of Ni-Mo-P/NF: (a) LSV curves; (b) overpotential η_{10}.

and Mo ions is insufficient, resulting in a decrease in the content of P and Mo in the coating. The hydrogen evolution catalytic performance of the coating is affected. The data obtained from the LSV detection method for calibration showed that when the $NiSO_4$ concentration is 0.16 mol/L, an electrode with better catalytic performance can be obtained.

RESULTS AND DISCUSSION

Through single-factor experiments, the optimal process parameters for preparing the Ni-Mo-P/NF ternary alloy hydrogen-evolving cathode were obtained. The electrodeposition solution was composed of 0.16 mol/L $NiSO_4$, 0.16 mol/L Na_2MoO_4, 0.20 mol/L NaH_2PO_2 and 0.25 mol/L $Na_3Cit{\cdot}2H_2O$, respectively. The electrodeposition should be conducted at the temperature of 55°C, pH 10, with current density of 10 A/dm^2. Continuous stirring was also carried out at 300 rmp/min. The Ni-Mo-P/NF electrode was prepared under the above conditions to research the phase morphology and the electrocatalytic performance.

Phase Morphology of the Ni-Mo-P Ternary Alloy Coating

Figure 5a, b and c shows the SEM images of the Ni-Mo-P/NF ternary alloy electrode surface enlarged at the different magnifications. Under the optimal process parameters, the prepared samples have a “honeycomb” morphology on the coating surface, which can provide a large specific reaction surface area and expose more active sites in the materials, thereby solving the problem of material transport and the discharge of hydrogen products. The EDS spectra given in Fig. 6 show that the three elements Ni, Mo and P are uniformly distributed in the Ni-Mo-P electrode (Table II).

Figure 7 shows the x-ray diffraction results of the Ni-Mo-P/NF alloy coating. In the x-ray diffraction

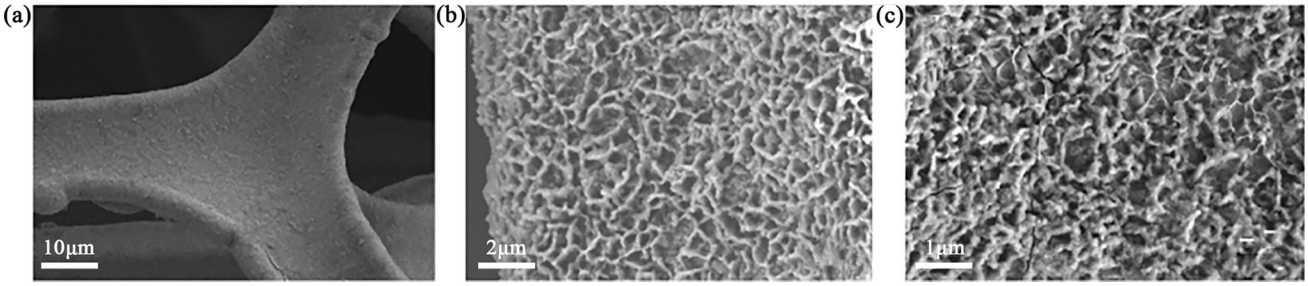

Fig. 5. SEM image of Ni-Mo-P/NF under the optimal process parameters. (a)–(c) represent electron microscope images at different magnification levels.

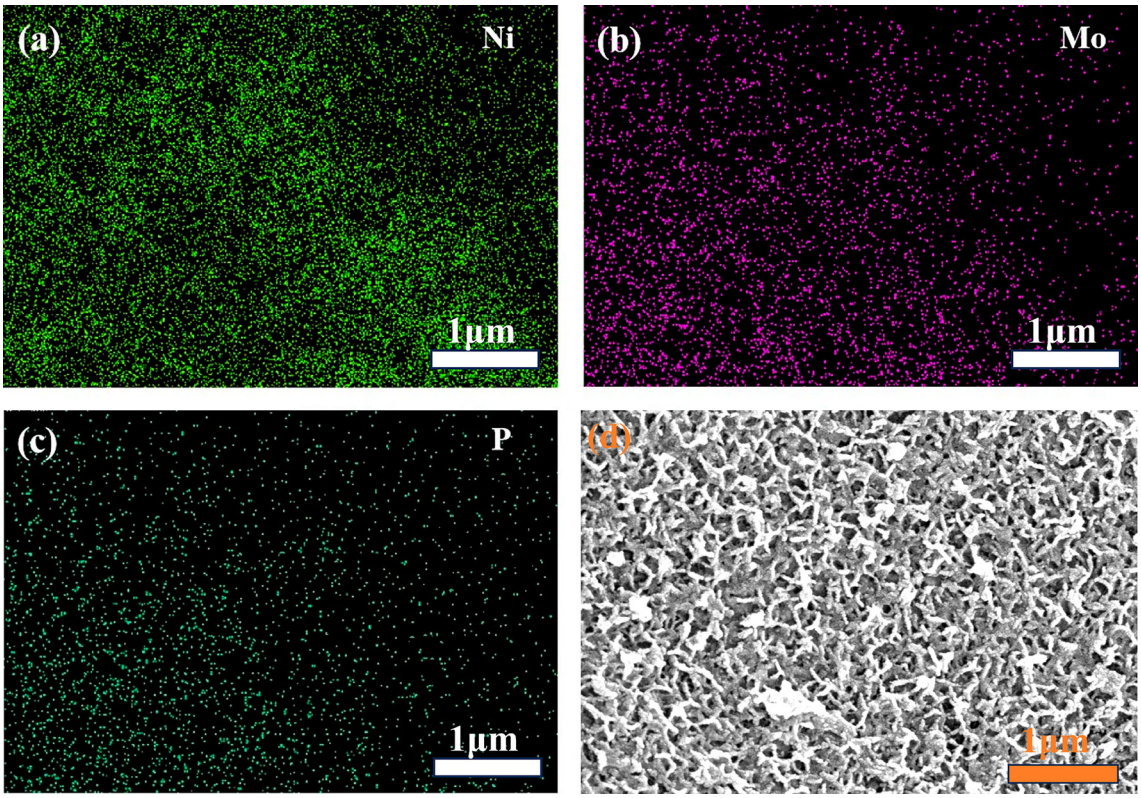

Fig. 6. Elemental distribution map of the Ni-Mo-P/NF electrode. (a) Distribution of the Ni element; (b) Distribution of the Mo element; (c) Distribution of the P element; (d) EDS electron image.

Table II. Spectrum diagram of the distribution map

Element	Wt%
Ni	46.58
Mo	32.79
P	20.63

pattern of Ni-Mo-P/NF alloy coating, no peak of P alone was observed, indicating that P does not exist alone but exists in a mixed form with Ni and Mo. For instance, Mo_3P on the (82) crystal plane and $MoNiP_2$ on the (111) crystal plane are presented. Furthermore, it was concluded that because the Mo_3P compound was generated, the reaction tendency of Mo with P is superior to that of Ni and P.

Analysis of the Electrocatalytic Performance of the Ni-Mo-P Ternary Alloy

The LSV results of the Ni-Mo-P/NF ternary porous electrode prepared under optimal conditions are shown in Fig. 8a. The catalytic hydrogen evolution performance of the porous electrode is better because its porous structure can increase the specific surface area of the electrode and effectively reduce the actual current density of the hydrogen

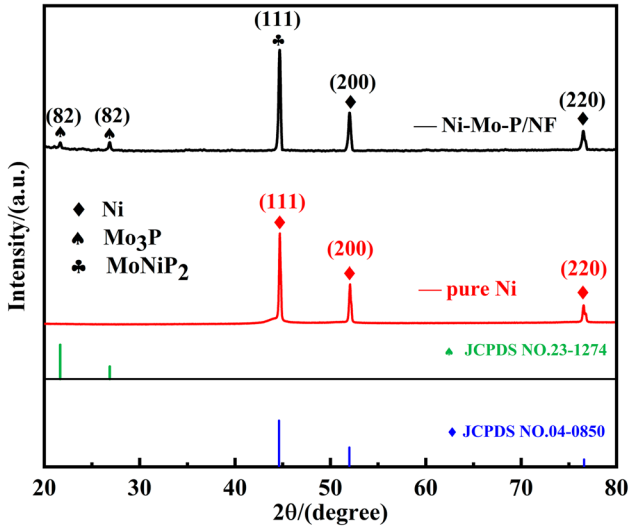

Fig. 7. X-ray diffraction pattern of the alloy electrode.

evolution reaction. Therefore, the NF with better hydrogen evolution catalytic performance was selected as the substrate. Using NF as the substrate, nickel, molybdenum and phosphorus are deposited electrochemically to achieve the goal of multicomponent alloying and structuring, that is, to modify the porous electrode with a highly active Ni-Mo-P coating to construct a long-lasting and stable, highly active porous composite hydrogen evolution cathode material. As shown in Fig. 8a, the Ni-Mo-P/NF ternary porous electrode has the lowest overpotential, η_{10} = 0.109 V, which means that it has relatively high electrocatalytic hydrogen evolution activity. Meanwhile, the Tafel slope of the Ni-Mo-P/NF catalytic electrode is 121.8 mV dec^{-1}, as shown in Fig. 8b. As shown in Fig. 8c, the charge transfer resistance of the Ni-Mo-P/NF electrode is 56.47 Ω. The Ni-Mo-P/NF electrode prepared under the optimal conditions has a lower resistance value.

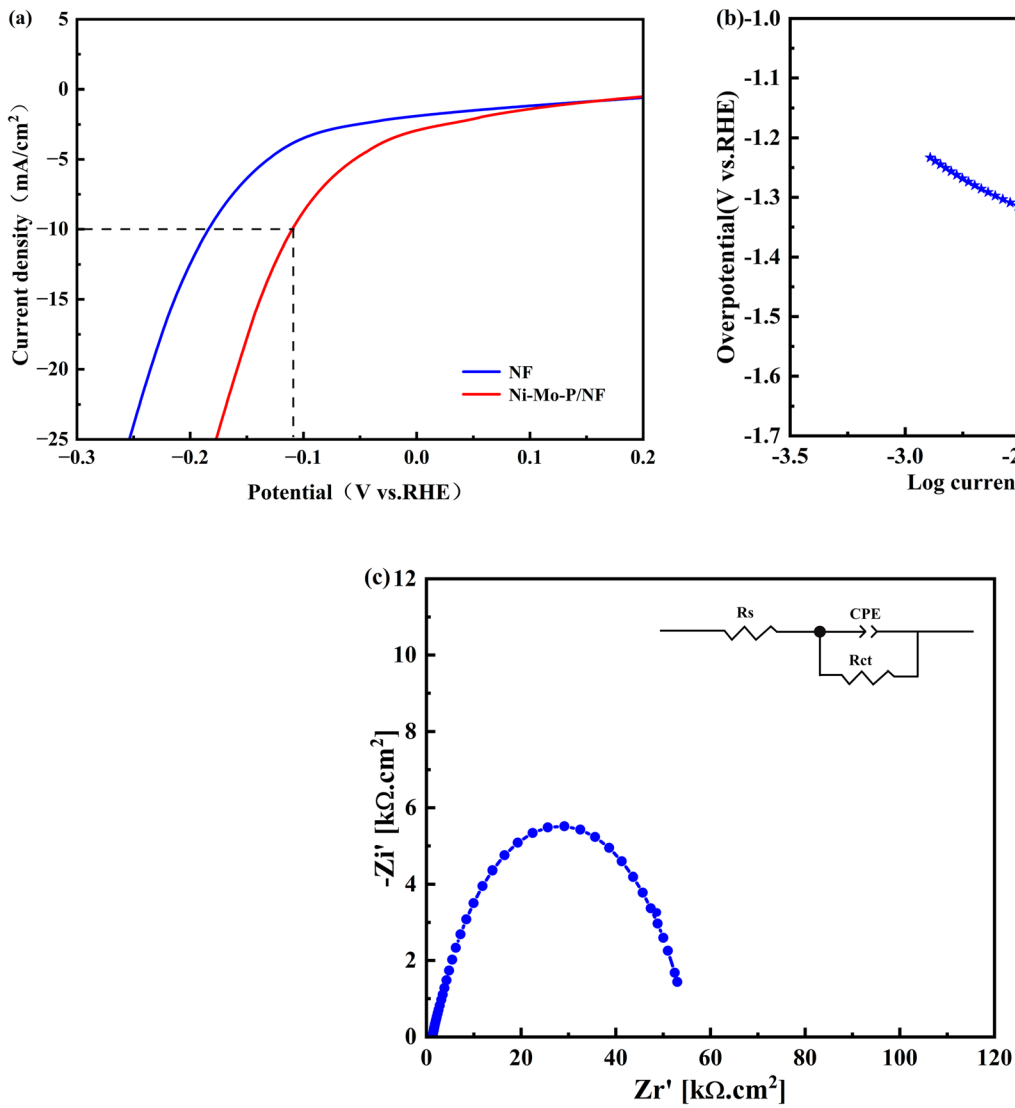

Fig. 8. (a) LSV for Ni-Mo-P/NF; (b) Tafel slope for Ni-Mo-P/NF; (c) EIS of the Ni-Mo-P/NF electrode.

Based on the above results, the Ni-Mo-P/NF hydrogen evolution cathode electrode has excellent HER catalytic activity and reaction kinetics.

When Cs = 0.04 mF/cm^2, ECSA = 87.6 cm^2 is obtained. The Ni-Mo-P/NF electrode has a relatively large electrochemically active specific surface area and many active sites, which are conducive to the hydrogen evolution reaction. A comparison of the LSV performance before and after the 3000-cycle CV test is shown in Fig. 9c. The electrode exhibited good stability, with just a slight decrease in activity after 3000-cycle CV. The overpotential only increased by 6 mV with a current density of 10 mA/cm^2.

Based on current density at an overpotential of 0.109 V for the electrodes obtained under optimal conditions, the number of active sites of Ni-Mo-P/NF was determined to be 1.7×10^{-8} mol by the ICP test, with the corresponding TOF = 0.31 s^{-1}. The above results indicate that the Ni-Mo-P/NF ternary alloy hydrogen evolution cathode electrode has good catalytic activity.

The comparison results for the main catalysts used it the present study are shown in Table III. The data indicate that the Ni-Mo-P/NF ternary alloy hydrogen evolution cathode electrode has a lower overpotential and lower energy consumption in practical applications and has certain practical value and technical and economic feasibility.

First-Principles Calculation

All simulation calculations were performed using the CASTEP module in Materials Studio. First, the original unit cell structure of the Ni-Mo binary alloy was established. A $2 \times 2 \times 2$ Ni phase crystal

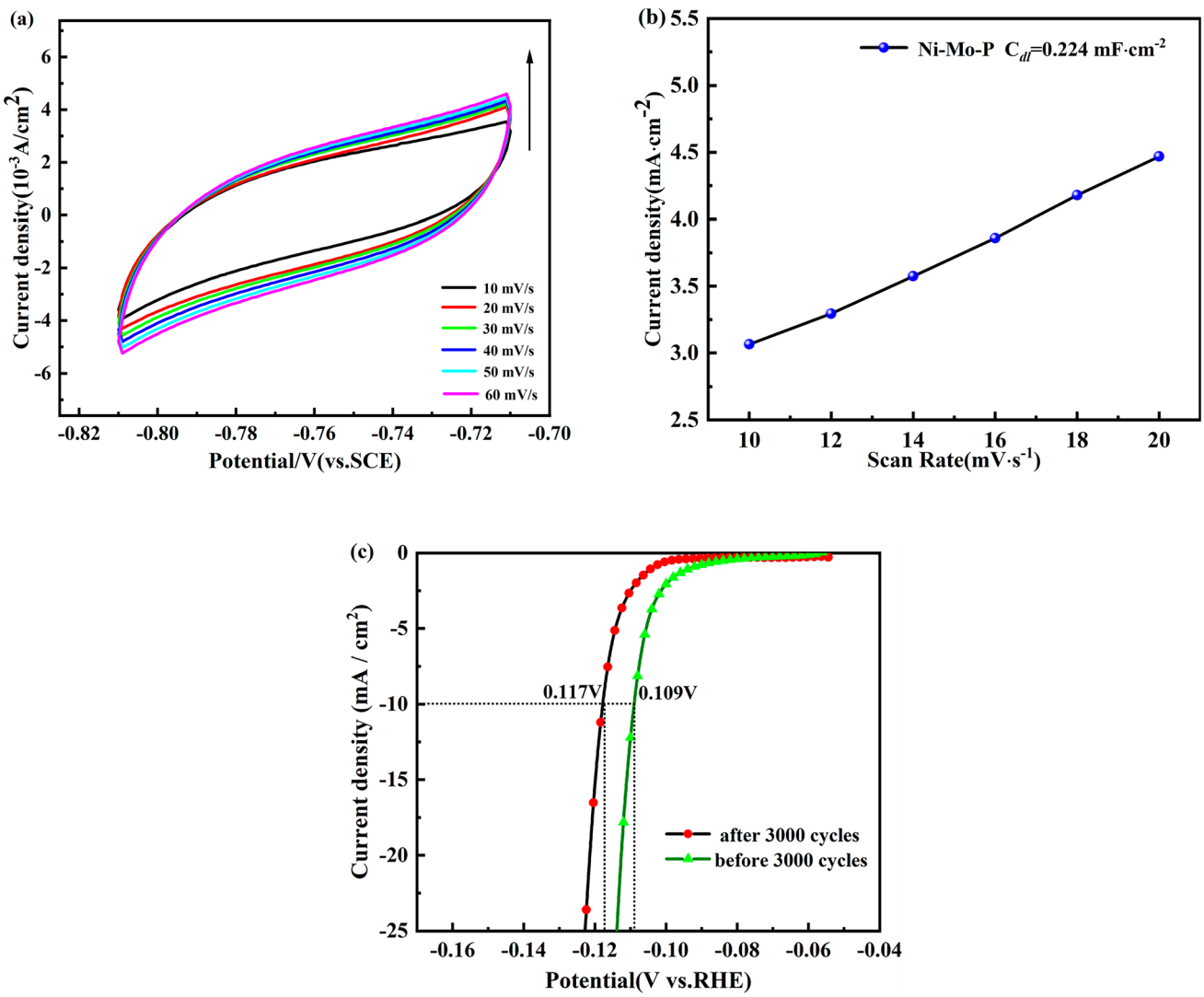

Fig. 9. (a) CV of Ni-Mo-P/NF electrodes; (b) C_{dl} of Ni-Mo-P/NF electrodes; (c) stability curves of Ni-Mo-P/NF electrodes after 3000 cycles CV scans.

Table III. Comparison of the HER performance of different hydrogen evolution catalysts

Electrode	Electrolyte	η_{10} (mV)	Tafel (mV/dec)	Ref
F-NCO	1 M KOH	236	125	40
$FeNiS_2$	1 M KOH	141	104	41
PyCoPc	1 M KOH	253	96	42
$NiFe_2O_4$/MXene	1 M NaOH	486	187.5	43
MoS_2	1 M KOH	160	46	44
Ni-Mo-P/NF	1 M KOH	109	121.8	This work

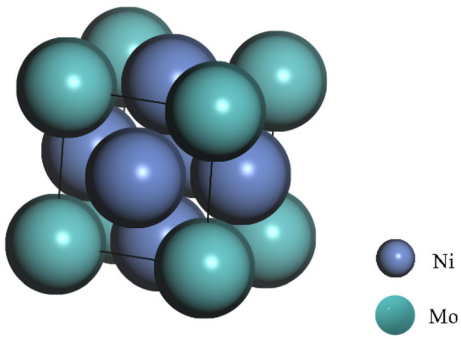

Fig. 10. Structural diagram of the Ni-Mo binary alloy.

structure model was selected. Within this crystal structure, by substituting a transition metal atom (Mo) for a Ni atom at the vertex of the cubic lattice, the Ni-Mo binary alloy structure was obtained, as shown in Fig. 10.

Using this model to calculate the formation energy of alloys, the structural stability of the Ni-Mo-P ternary alloy was compared and analyzed. The calculation formula is shown in Eq. 11:

$$\Delta H_{alloy} = E(Ni_{24/32}Mo_{7/32}M_{1/32}, a) - 24E_{Ni}(a^0_{Ni}) - 7E_{Mo}(a^0_{Mo}) - E_M(a^0_M) \tag{11}$$

In the formula $E(Ni_{24/32}Mo_{7/32}M_{1/32}, a)$, the total energy of the alloy structure includes 24 Ni atoms, 7 Mo atoms and 1 P atom; $E_{Ni}(a^0_{Ni})$, $E_{Mo}(a^0_{Mo})$ and $E_M(a^0_P)$ are the total energies of a Ni atom, a Mo atom and a P atom in their ground states, respectively. The formation energy of the alloy, calculated by the first-principles method, is more negative, indicating that the energy form of this structure is more stable.

Based on the Ni-Mo binary alloy structure, the surface with the highest reaction probability in the binary alloy structure model was subjected to surface cutting to obtain a surface structure composed of five atomic layers. A vacuum with a thickness of 1.3 nm was established, with the bottom two layers fixed and the surface three layers freely relaxed, as shown in Fig. 11a. The Ni and Mo atoms in the second and third layers of the surface structure crystal cell in Fig. 11a were replaced by P atoms to obtain the structure model shown in Fig. 11b. The models of H* adsorption on the crystal planes of each material were established, as shown in Fig. 11c.

The surface structure of the Ni-Mo-P modified coating obtained above was used for the calculation. The calculation formula is as follows:

$$E_H = E_{surface-H} - E_{surface} - \frac{1}{2}E_{H_2} \tag{12}$$

Based on the calculation in the previous text, the adsorption energy of hydrogen ΔE_H is obtained. By combining Eqs. 13, 14 and 15, the free energy ΔG_{H*} of the adsorbed hydrogen can be transformed from the adsorption energy of hydrogen. The formula is as follows:

$$\mathrm{ZPE} = \sum_i \frac{hv_i}{2} \tag{13}$$

$$S = K_B \sum_i \left(\frac{x_i}{e^{x_i} - 1} - In(1 - e^{-x_i})\right) \tag{14}$$

$$\Delta G_{H*} = \Delta E_H + \Delta E_{ZPE} - T\Delta S_H \tag{15}$$

Furthermore, the adsorption free energy (ΔG_{H*}) of the Ni-Mo-P ternary alloy was compared. The adsorption free energy of hydrogen by an ideal catalyst should be close to zero to achieve a balance of adsorption and desorption efficiency. In the Ni-Mo-P catalytic material, the metal sites act as active sites for adsorbing H* and H_2O. Through density functional theory (DFT) calculations, the catalytic effect of Ni (111) surface d orbitals on H_2O dissociation was revealed. The hybridization of the d electron orbitals of Ni with the p orbitals of the O atom of H_2O reduced the energy barrier for H-O bond breakage (Volmer step). The doping of Mo and P induced lattice distortion of Ni, resulting in the formation of more defects and edge sites, thereby increasing the intrinsic activity. The Gibbs free energy of hydrogen adsorption of the Ni-Mo-P ternary alloy material is − 0.432 eV, as shown in Fig. 12.

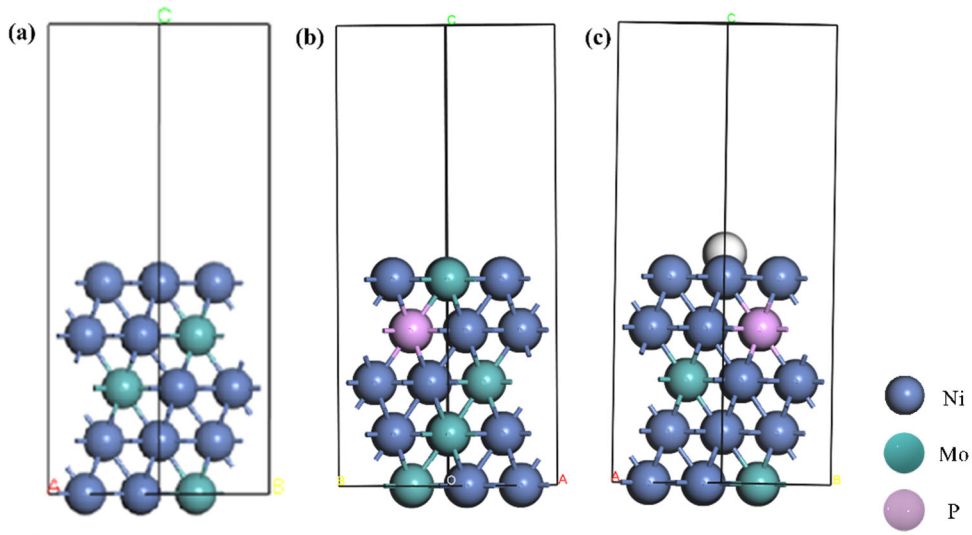

Fig. 11. Crystal cell model of the Ni-Mo coating surface; structural diagram of the ternary alloy Ni-Mo-P; schematic diagram of the calculation model for the hydrogen adsorption energy. (a) Crystal cell model of the Ni-Mo coating surface; (b) structural diagram of the ternary alloy Ni-Mo-P; (c) schematic diagram of the calculation model for the hydrogen adsorption energy.

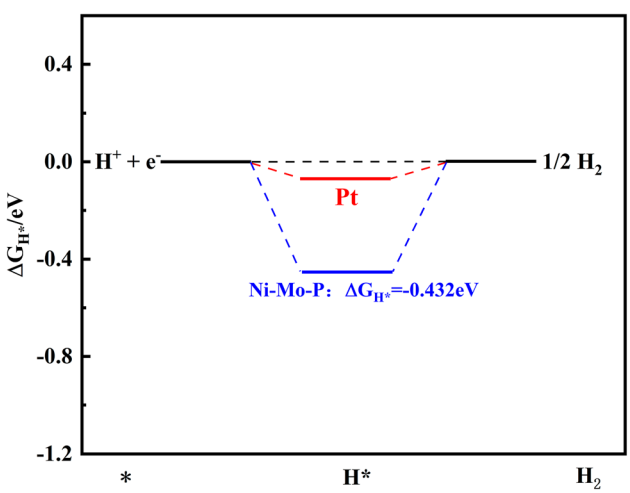

Fig. 12. Schematic diagram of the adsorption free energy of the Ni-Mo-P ternary alloy.

CONCLUSIONS

In this paper, the electro-deposition method was adopted to conduct single-factor experiments to explore the optimal composition of the electrolyte for preparing Ni-Mo-P/NF hydrogen-evolving cathode. Morphology characterization and electrochemical performance tests were also conducted. Combined with first-principles calculations, the hydrogen-evolving reaction mechanism of the Ni-Mo-P/NF ternary alloy hydrogen-evolving cathode electrode is revealed.

(1) The first-principles calculation results indicate that the introduction of Mo and P elements alters the electronic structure of Ni. The d orbitals of Mo hybridize with those of Ni. Meanwhile, the introduction of P atoms enhanced the charge transfer ability at the electrode surface and improved the catalytic activity of the Ni-Mo-P/NF ternary alloy hydrogen evolution cathode electrode.

(2) The optimal composition of the plating solution and process parameters are as follows: the $NiSO_4$ concentration is 0.16 mol/L, the Na_2MoO_4 concentration is 0.16 mol/L, the NaH_2PO_2 concentration is 0.20 mol/L, the $Na_3Cit{\cdot}2H_2O$ concentration is 0.25 mol/L, the current density is 10 A/dm^2, the deposition time is 50 min, the temperature is 55°C, pH is 10, and the stirring speed is 300 rpm/min. The surface of the ternary coating presented a "honeycomb" structure, affording many active chemical reaction active sites, a high charge transfer efficiency and fast hydrogen release rate. The XRD results show that amorphous solid solutions of Mo_3P and $MoNiP_2$ exist during the deposition process.

(3) Under the test conditions of 1 M KOH solution at room temperature, the overpotential η_{10} for hydrogen evolution of the Ni-Mo-P/NF porous hydrogen evolution electrode was 0.109 V; the Tafel slope was 121.8 mV/dec; the AC impedance Rct was 56.47 Ω; the specific active surface area was 87.6 cm^2; the change in overpotential before and after 3000 CV cycles of the Ni-Mo-P/NF porous hydrogen evolution electrode was $< 10\%$, indicating good stability.

ACKNOWLEDGMENTS

This work was supported by the Regional Innovation Capability Guidance Plan of Shaanxi Province (grant no. 2022QFY10-05), Key Projects with Provincial and Municipal Collaboration of Shaanxi Province (grant nos. 2022GD-TSLD-17, 2022GD-TSLD-11), "Double-chain" school-enterprise key projects of Shaanxi Province (grant no. 2022LL-JB-09) and the Scientific and Technological Innovation Team Project of Shaanxi Innovation Capability Support Plan (grant no. 2022TD-30).

FUNDING

Natural Science Basic Research Program of Shaanxi Province, 2022QFY10-05, Jinjing Du, Key Projects with Provincial and Municipal Collaboration of Shaanxi Province, 2022GD-TSLD-17, Jinjing Du, 2022GD-TSLD-11, Jinjing Du, Key Research Institute of Humanities and Social Sciences in Sichuan Province, 2022LL-JB-09, Jinjing Du, Program for Jiangsu Provincial Excellent Scientific and Technological Innovation Team, 2022TD-30, Jinjing Du

CONFLICT OF INTEREST

On behalf of all the authors, the corresponding author states that there are no conflicts of interest.

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